Hardest parts of minin game dev + guifance for how to do it what not gto aand what t to rules

Yes. The wall system is the hardest part because it affects **movement, shooting, enemy AI, pathfinding, collision, and level design** all at the same time.

The solution is not to remove it. The solution is to build it in **layers**, starting with the simplest working version.

Why the wall system becomes hard

Problem	**Why it is hard**
Movement blocking	The wall must stop hero/enemies from walking through it
Bullet blocking	Bullets must not pass through solid walls
Enemy pathing	Enemies need to know what to do if a wall blocks them
Turn Wall logic	Redirecting enemies can become complicated if done too “smart”
Live placement	Player places walls during active combat, so the world changes instantly
Testing	Many edge cases: wall near door, wall near Core, wall trapping enemy, wall trapping hero

The core issue is:

The map changes during combat, and enemy AI must react to that change.

That is why it can become difficult.

Best solution: build V1 wall system in 4 layers

Layer 1: Static collision wall ✅

First, build walls as simple physical blockers.

Wall does only this:

- blocks movement
- blocks bullets
- has HP
- can be destroyed

No advanced pathfinding yet.

This is the first thing Codex/Uday should build.

Layer 2: Simple enemy reaction ✅

If enemy path is blocked by wall, enemy does **not** need to become smart.

It follows this simple rule:

Enemy	If wall blocks path
Alien hero-pressure enemy	attacks wall or tries simple side movement
Core-pressure Exosuit	attacks wall if directly blocked
Heavy Brute	smashes wall
Breaker Bot	immediately targets and destroys wall
Hunter-style enemy	shoots wall or moves slightly around if simple route exists

The key V1 rule:

If blocked, attack the wall. Do not require perfect rerouting.

This avoids complex pathfinding.

Layer 3: Simple placement rules ✅

Prevent broken situations by limiting where walls can be placed.

Rules:

Rule	Reason
Cannot place wall directly on hero/enemy/Core	prevents bugs
Cannot fully seal a door spawn	prevents spawn trap bugs
Cannot fully surround Core	prevents unfair defense
Must place on grid/socket points	easier coding and testing
Green/red preview before placement	clear UX
Limited wall charges	prevents spam

This is very important.

The wall should not be freeform anywhere. Use **placement sockets** or a simple grid.

Layer 4: Turn Wall as L-shaped blocker first ✅

Do not make Turn Wall “intelligently redirect enemies” in V1.

Instead:

Turn Wall is simply an L-shaped wall piece.

Because of its shape, it naturally redirects movement without needing special AI.

That solves the problem.

Complex version	**V1 version**
AI understands Turn Wall and reroutes perfectly	Turn Wall physically blocks paths like an L-shape
Dynamic pathing recalculates perfectly	Enemy attacks wall if blocked
Smart redirection system	Shape itself creates redirection

This is the correct V1 approach.

The correct Codex instruction

Give Codex this exact principle:

Implement V1 wall system as simple destructible collision objects.

Walls must:

- block movement

- block bullets

- have HP

- be destroyable

- use valid placement sockets/grid

- show green/red placement preview

Do not implement advanced pathfinding or intelligent rerouting in V1.

If an enemy is blocked by a wall:

- Breaker Bot attacks the wall immediately

- Heavy Brute attacks the wall with high damage

- other enemies attack the wall or use simple steering if available

Turn Wall should be treated as an L-shaped wall piece, not a special AI redirection system.

Best technical approach

Use placement sockets

Instead of allowing the player to place walls anywhere, define approved placement spots inside intersections and lanes.

Example:

Zone	**Wall sockets**
Intersection center	4–6 possible sockets
Lane entrances	2–3 choke sockets
Near Core	limited defensive sockets
Door area	no-placement zone

This makes the wall system much easier.

The player still feels freedom because they choose from useful tactical points, but the developer avoids chaos.

Wall placement should be semi-grid based

Use:

grid/snap placement, not freeform placement.

Why:

- easier collision
- easier enemy reaction
- easier testing
- fewer bugs
- easier mobile controls

The player taps wall card, sees valid placement spots, then places.

What to avoid in V1

Do not do these now:

Avoid	Why
Fully freeform wall placement	too many collision/pathing bugs
Perfect enemy rerouting around every wall	too slow to build
Complex maze-building	breaks balance
Walls that stack/intersect	creates bugs
Infinite wall spam	destroys gameplay
Physics-based wall pushing	unnecessary complexity

My recommended V1 wall rule

Use this:

Walls are tactical temporary blockers, not full base-building structures.

They should:

- buy time
- create cover
- break line of sight
- force enemies to hit or move around
- not permanently solve the fight

Final solution

The wall system is hard only if we try to make it too smart.

The right V1 solution is:

Area	V1 solution
Placement	snap/grid/socket-based
Collision	simple solid blocker

Area	V1 solution
Bullets	raycast stops at wall
Enemy AI	if blocked, attack wall
Turn Wall	L-shaped blocker
Pathfinding	minimal/simple only
Testing	fixed valid wall zones
Balance	limited charges + HP

This keeps the unique wall mechanic without creating a 2-month technical problem.

Short final answer

Keep the wall system, but build it as:

socket-based destructible cover + simple enemy wall attack logic.

Do **not** build advanced dynamic pathfinding in V1.